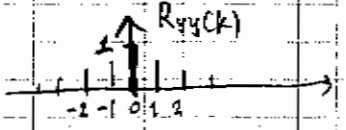
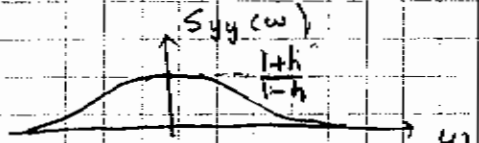
Problem 4.1

1)





2) $R_{yy}(k) = E[Y(n)Y(n+k)]$

$= E[Y(n)(h \cdot Y(n+k-1) + Z(n+k))]$

$= h \cdot R_{yy}(k-1)$ ok

3) $\begin{cases} R_{yy}(0) = h^2 R_{yy}(0) + \sigma_Z^2 \\ R_{yy}(0) = 1 \end{cases}$

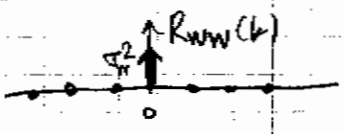
$\Downarrow$

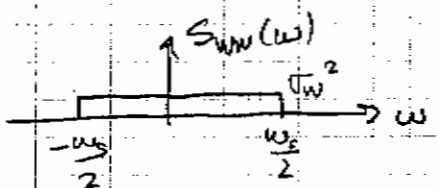
$\sigma_Z^2 = 1 - h^2 = 0{,}36$

4) $H_{gen}(z) = \frac{Y(z)}{Z(z)} = \frac{1}{1 - hz^{-1}}$

5) se m-fil

6)





7) se m-fil

8) $R_{xx}(k) = E[x(n)x(n+k)]$

$= E[(Y(n)+W(n))(Y(n+k)+W(n+k))]$

$= R_{yy}(k) + R_{ww}(k) + 0 + 0$

$S_{xx}(z) = S_{yy}(z) + S_{ww}(z)$

9) $R_{xy}(k) = E[x(n)\,y(n+k)]$

$= E[(y(n)+W(n))\,y(n+k)]$

$= R_{yy}(k)$

$S_{xy}(z) = S_{yy}(z)$

10) $MSE_X = E[W^2(n)] = \sigma_W^2 = 4$

11) se m-fil

12) $$\begin{bmatrix} R_{yy}(0)+\sigma_w^2 & R_{yy}(1) \\ R_{yy}(1) & R_{yy}(0)+\sigma_w^2 \end{bmatrix}\begin{bmatrix} h_0 \\ h_1 \end{bmatrix} = \begin{bmatrix} R_{yy}(0) \\ R_{yy}(1) \end{bmatrix}$$

$$\Downarrow$$

$$\begin{bmatrix} 1+\sigma_w^2 & h \\ h & 1+\sigma_w^2 \end{bmatrix}\begin{bmatrix} h_0 \\ h_1 \end{bmatrix} = \begin{bmatrix} 1 \\ h \end{bmatrix}$$

$$\Downarrow$$

$$\begin{bmatrix} h_0 \\ h_1 \end{bmatrix} = \frac{1}{(1+\sigma_w^2)^2-h^2}\begin{bmatrix} 1+\sigma_w^2-h^2 \\ h\sigma_w^2 \end{bmatrix} = \begin{bmatrix} 0,1790 \\ 0,1314 \end{bmatrix}$$

13) $$MSE_{\hat{y}} = R_{yy}(0) - \sum_0^M h(m)R_{xy}(m)$$

$$= \sigma_w^2 \frac{1+\sigma_w^2-h^2}{(1+\sigma_w^2)^2-h^2}$$

$$= \underline{0,7159} \qquad (\ll MSE_x)$$

14) $$SNR_{in} = \frac{P_y}{P_w} = \frac{R_{yy}(0)}{\sigma_w^2} = 1/4$$

$$SNR_{out} = \frac{P_{\hat{y}}}{P_{w_{out}}} = \frac{\frac{T_s}{2\pi}\int_{-\omega_s/2}^{\omega_s/2} S_{yy}(\omega)|H(\omega)|^2 d\omega}{\frac{T_s}{2\pi}\int_{-\omega_s/2}^{\omega_s/2} \sigma_w^2 |H(\omega)|^2 d\omega}$$

15) se m-fil

16) se m-fil $M \to \infty$ ~ causalt IIR Wiener filter

17) $$\begin{bmatrix} 1+\sigma_w^2 & h & h^2 \\ h & 1+\sigma_w^2 & h \\ h^2 & h & 1+\sigma_w^2 \end{bmatrix}\begin{bmatrix} h_{-1} \\ h_0 \\ h_1 \end{bmatrix} = \begin{bmatrix} h \\ 1 \\ h \end{bmatrix}$$

$$\Downarrow$$

$$\begin{bmatrix} h_{-1} \\ h_0 \\ h_1 \end{bmatrix} = \frac{1}{(1+\sigma_w^2+h^2)(1+\sigma_w^2)-2h^2}\begin{bmatrix} h\sigma_w^2 \\ 1+\sigma_w^2-h^2 \\ h\sigma_w^2 \end{bmatrix} = \begin{bmatrix} 0,1190 \\ 0,1622 \\ 0,1190 \end{bmatrix}$$

Lineær fase pga. medført symmetri i $R_{yy}(k)$

18) $$MSE_3 = R_{yy}(0) - \sum_{-1}^{1} h(m)R_{xy}(m) = 1-2hh_1-h_0 = \underline{0,6474}$$

19) se m-fil

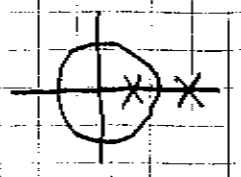
20) — h — $M \to \infty$ svarer til non-causal IIR

21) Non-causal FIR bedre end causal FIR af samme længde pga. $R_{yy}(k)$

22) $H(z) = \frac{S_{xy}(z)}{S_{xx}(z)} = \frac{S_{yy}(z)}{S_{yy}(z)+S_{ww}(z)}$

$= \frac{1-h^2}{1-h^2+\sigma_w^2(1-hz^{-1})(1-hz)}$

$= \frac{0,07531}{(1-0,6691z^{-1})(1-0,6691z)}$

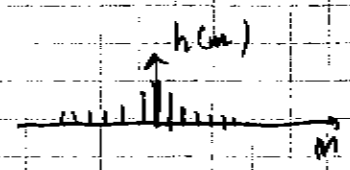


23) Kan realiseres stabilt XOR causalt

⇓ [Stabilt ⇒ non-causalt]

24) $h(m) = \mathcal{Z}^{-1}\{H(z)\}$

$= 0,1364 \cdot (0,6691)^{|m|}$



25) $MSE_S = R_{yy}(0) - \sum_{-\infty}^{\infty} h(m) R_{xy}(m)$

$= 1 - \sum_{-\infty}^{\infty} 0,1364 \cdot (0,6691)^{|m|} \cdot (0,8)^{|m|}$

$= 1 - 0,1364 \sum_{-\infty}^{\infty} (0,5353)^{|m|}$

$= 1 - 0,1364 \left(2 \frac{1}{1-0,5353} - 1\right)$

$= \underline{0,5494}$

26) $MSE_S = R_{yy}(0) - \sum_{-1}^{1} h(m) R_{xy}(m)$

$= \underline{0,7176}$

27) se m-fil

28) Non-causalt FIR bedst

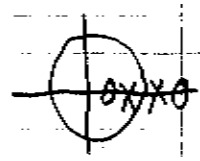
29) $S_{xx}(z) = S_{yy}(z) + S_{ww}(z)$

$$= \frac{1-h^2}{(1-hz^{-1})(1-hz)} + \sigma_w^2$$

$$= \frac{1-h^2+\sigma_w^2(1+h^2) - h\sigma_w^2(z+z^{-1})}{(1-hz^{-1})(1-hz)}$$

$$= \frac{6{,}92 - 3{,}2(z+z^{-1})}{(1-0{,}8z^{-1})(1-0{,}8z)}$$

$$= 4{,}7802 \frac{(1-0{,}6691z^{-1})(1-0{,}6691z)}{(1-0{,}8z^{-1})(1-0{,}8z)}$$



30) $$= \underbrace{2{,}1864 \cdot \frac{1-0{,}6691z^{-1}}{1-0{,}8z^{-1}}}_{G^+(z)} \cdot \underbrace{2{,}1864 \frac{1-0{,}6691z}{1-0{,}8z}}_{G^-(z)}$$

31) $$G_{White}(z) = \frac{1}{G^+(z)} = 0{,}4574 \frac{1-0{,}8z^{-1}}{1-0{,}6691z^{-1}}$$

32) $$H_{MSE}(z) = \text{causal}\left\{\frac{S_{yy}(z)}{G^-(z)}\right\}$$

$$= \text{causal}\left\{0{,}1647 \frac{1}{(1-0{,}8z^{-1})(1-0{,}6691z)}\right\}$$

$$= \text{causal}\left\{\frac{0{,}1647\, z^{-1}}{(1-0{,}8z^{-1})(z^{-1}-0{,}6691)}\right\}$$

$$= \text{causal}\left\{\frac{0{,}3544}{1-0{,}8z^{-1}} + \frac{0{,}2371}{z^{-1}-0{,}6691}\right\}$$

$$= \frac{0{,}3544}{1-0{,}8z^{-1}}$$

33) $$H_{Wiener}(z) = G_{White}(z)\, H_{MSE}(z) = \frac{0{,}1621}{1-0{,}6691\,z^{-1}}$$

$$h(m) = 0{,}1621\,(0{,}6691)^m u(m) \quad \text{causal AND stabil}$$

34) $$MSE_3 = R_{yy}(0) - \sum_0^\infty h(m) R_{xy}(m)$$

$$= 1 - 0{,}1621 \frac{1}{1-0{,}8 \cdot 0{,}6691}$$

$$= \underline{0{,}6512}$$

35) se m-fil

36) teoretisk bedst = IIR non-causal
= FIR non-causal ($M \to \infty$)

bedst endeligt non-causalt = FIR non-causalt
bedre end IIR trunkeret

bedst causalt = IIR causalt
= FIR causalt ($M \to \infty$)

for given filterlgd $< \infty$ vælg FIR

ooo

Problem 4.2

1) $R_{xy}(k) = E[x(n)\,y(n+k)]$

$= E[[v(n)+w(n)]\,y(n+k)]$

$= R_{vy}(k)$

$= g(-k) * R_{yy}(k)$

$S_{xy}(z) = G^*(z)\,S_{yy}(z)$

2) $S_{xx}(f) = |G(f)|^2 S_{yy}(f) + S_{ww}(f)$ eller

$S_{xx}(z) = G(z)G^*(z)S_{yy}(z) + S_{ww}(z)$

3–7) se Problem (13+14) – 1

8) se m-fil

9–10) se Problem (13+14) – 1

11) se m-fil

12) $S_{xy}(z) = G^*(z)\,S_{yy}(z)$

$= \dfrac{1-h^2}{(1-hz^{-1})(1-hz)(1-bz)}$

$= \dfrac{0{,}9\,z^{-1}}{(1-0{,}8z^{-1})(1-1{,}25z^{-1})(1-2z^{-1})}$

$= \dfrac{1}{1-2z^{-1}} + \dfrac{-8/3}{1-1{,}25z^{-1}} + \dfrac{5/3}{1-0{,}8z^{-1}}$

$= \dfrac{-\frac{1}{2}z}{1-\frac{1}{2}z} + \dfrac{32/15\,z}{1-0{,}8z} + \dfrac{5/3}{1-0{,}8z^{-1}}$

13) $R_{xy}(k) = -\frac{1}{2}2^{k+1}u(-(k+1)) + \frac{32}{15}\left(\frac{5}{4}\right)^{k+1}u(-(k+1)) + \frac{5}{3}\left(\frac{4}{5}\right)^{k}u(k)$

$= -2^k u(-n-1) + \frac{8}{3}\left(\frac{5}{4}\right)^k u(-n-1) + \frac{5}{3}\left(\frac{4}{5}\right)^k u(k)$

$= \begin{cases} \frac{5}{3}\left(\frac{4}{5}\right)^k & k \geq 0 \\ \frac{8}{3}\left(\frac{5}{4}\right)^k - 2^k & k < 0 \end{cases}$

14) $S_{xx}(z) = G(z)G^*(z)S_{yy}(z) + S_{ww}(z)$

$$= \frac{1-h^2}{(1-hz^{-1})(1-hz)(1-bz^{-1})(1-bz)} + \sigma_w^2$$

$$= 4 + \left(-\frac{4}{3}\right)\frac{3/4}{(1-0{,}5z^{-1})(1-0{,}5z)} + \frac{40}{9}\frac{0{,}36}{(1-0{,}8z^{-1})(1-0{,}8z)}$$

15) $R_{xx}(k) = 4\delta(k) + \frac{40}{9}(0{,}8)^{|k|} - \frac{4}{3}(0{,}5)^{|k|}$

16) $\begin{bmatrix} 5/3 \\ 4/3 \end{bmatrix} = \begin{bmatrix} 64/9 & 26/9 \\ 26/9 & 64/9 \end{bmatrix}\begin{bmatrix} h_0 \\ h_1 \end{bmatrix}$

$$\Downarrow$$

$$\begin{bmatrix} h_0 \\ h_1 \end{bmatrix} = \begin{bmatrix} 0{,}1895 \\ 0{,}1105 \end{bmatrix}$$

17) $MSE_{\downarrow} = R_{yy}(0) - \sum_{0}^{n} h(m) R_{xy}(m)$

$= 1 - h_0 R_{xy}(0) - h_1 R_{xy}(1)$

$= \underline{0{,}5368}$

18) 19) se m-fil

20) $\begin{bmatrix} 49/30 \\ 5/3 \\ 4/3 \end{bmatrix} = \frac{1}{9}\begin{bmatrix} 64 & 26 & 113/5 \\ 26 & 64 & 26 \\ 113/5 & 26 & 64 \end{bmatrix}\begin{bmatrix} h_{-1} \\ h_0 \\ h_1 \end{bmatrix}$

$$\Downarrow$$

$$\begin{bmatrix} h_{-1} \\ h_0 \\ h_1 \end{bmatrix} = \begin{bmatrix} 0{,}1434 \\ 0{,}1443 \\ 0{,}0782 \end{bmatrix}$$

$R_{xy}(k)$ ikke symmetrisk $\Rightarrow$ ikke lineær fase

21) $MSE_{\downarrow} = R_{yy}(0) - \sum_{-n}^{n} h(m) R_{xy}(m)$

$= 1 - \sum_{-1}^{1} h(m) R_{xy}(m)$

$= \underline{0{,}4209}$

22) 23) se m-fil

24) Non-causalt bedst pga. store korrelation